Question Paper

Exam Date & Time: 25-Nov-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS & COMMUNICATION ENGINEERING) DEGREE EXAMINATIONS - NOVEMBER / DECEMBER 2024
SUBJECT: ECE 2123/ECE_2123 - SIGNALS & SYSTEMS

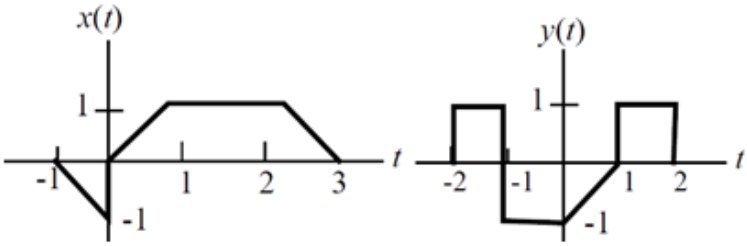
Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) A stable system has the indicated input $x(t)$ and output $y(t)$. Use Laplace transforms to determine the transfer function and impulse response of the system. (4)

(i). $x(t) = e^{-t}u(t), y(t) = e^{-2t}\cos(t)\,u(t)$
(ii). $x(t) = e^{-2t}u(t), y(t) = -2e^{-t}u(t) + 2e^{-3t}u(t)$

1B) Let $x(t)$ and $y(t)$ are given below: (3)



Sketch the following signals:

(i). $x(t)y(t-1)$
(ii). $x(t)y(2-t)$
(iii). $x(2t)y(\frac{1}{2}t+1)$

1C) (3)

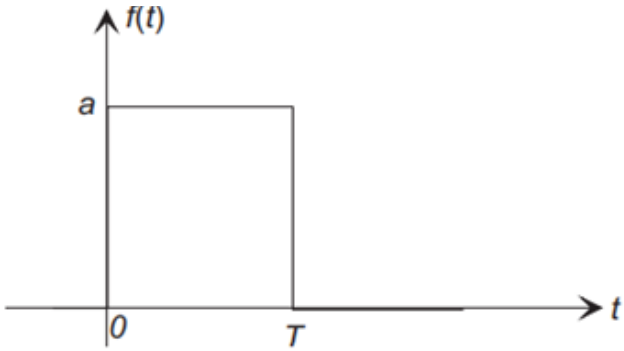
Given $x[n] = \frac{\sin\left(\frac{11\pi}{20}n\right)}{\sin\left(\frac{\pi}{20}n\right)} \xleftrightarrow{DTFS} X[k]$, evaluate the time domain signal $y[n]$ with the following DTFS coefficients, using only DTFS properties:

(i). $Y[k] = X[k-5] + X[k+5]$
(ii). $Y[k] = cos\left(k\frac{\pi}{5}\right)X[k]$
(iii). $Y[k] = X[k] \circledast X[k]$

2A) (4)

Determine the zero state and zero output response for the system described by $y[n] + (\frac{1}{3})y[n-1] = x[n]$, for the input $x[n] = \left(\frac{1}{2}\right)^n u[n]$ with $y[-1] = 1$.

2B) Find the autocorrelation function of the square pulse of amplitude a and duration T as shown below. Also find and sketch the energy spectral density of the signal. (3)



2C) Determine and sketch the convolution of the following two signals- (3)

$$x(t) = \begin{cases} t+1, 0 \le t \le 1 \\ 2-t, 1 \le t \le 2 \\ 0, elsewhere \end{cases}$$

$$h(t) = \delta(t+2) + 2\delta(t+1)$$

3A) Using DTFT, determine the frequency response and hence impulse response of the system defined by the following difference equation: (4)
y[n−2] + 5y[n−1] + 6y[n] = 8x[n−1] + 18x[n].

3B) An LTI system has impulse response $h(t) = \frac{1}{\pi t}\sin(\pi t)$. The input signal shown in Fig. 3B is applied to the system. Use FT to determine the output of the system. (3)

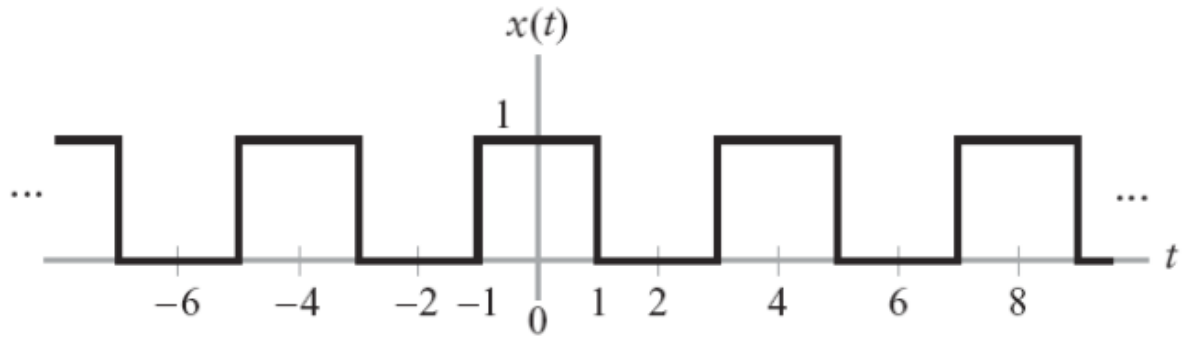


Fig. 3B

3C) Evaluate the convolution: $y[n] = b^n u[n] * a^n u[n-10]$ where, $|a| < 1$ and $|b| < 1$, for (i) $a \neq b$, (ii) $a = b$. (3)

4A) Consider a discrete-time system depicted in Fig. 4A. Use DTFT to determine the output y[n] if x[n], w[n] and h[n] are as below. Also sketch $G(e^{j\Omega})$. (4)

$$x[n] = h[n] = \frac{\sin(\pi n/2)}{\pi n}, \quad w[n] = \cos(\pi n/2)$$

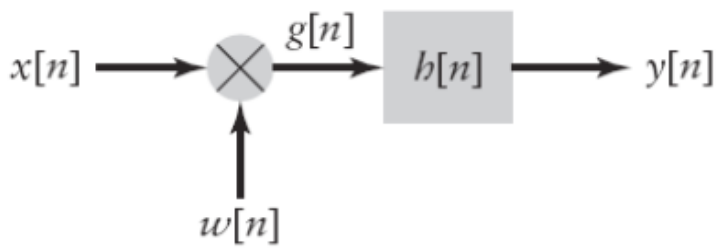


Fig. 4A

4B) Signals x[n] and y[n] are depicted in Fig. 4B. (3)
Sketch (i) x[4 – n]y[0.5n – 2], (ii) x[2n] + 0.5y[n]

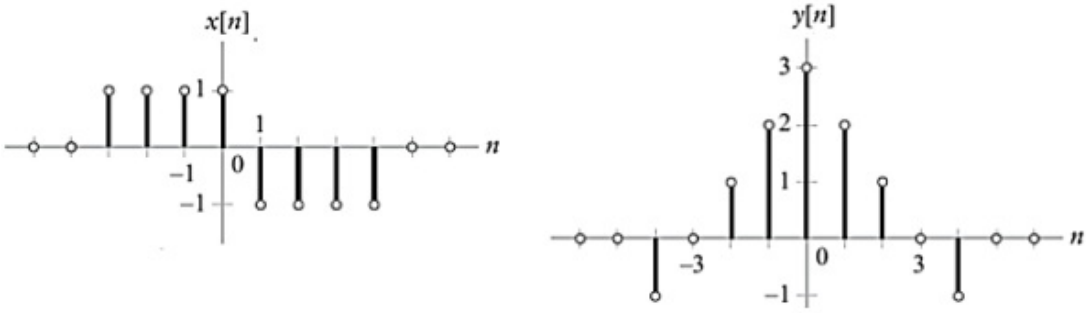


Fig. 4B

4C) Determine the power spectral density of the periodic signal shown in Fig. 4C. (3)

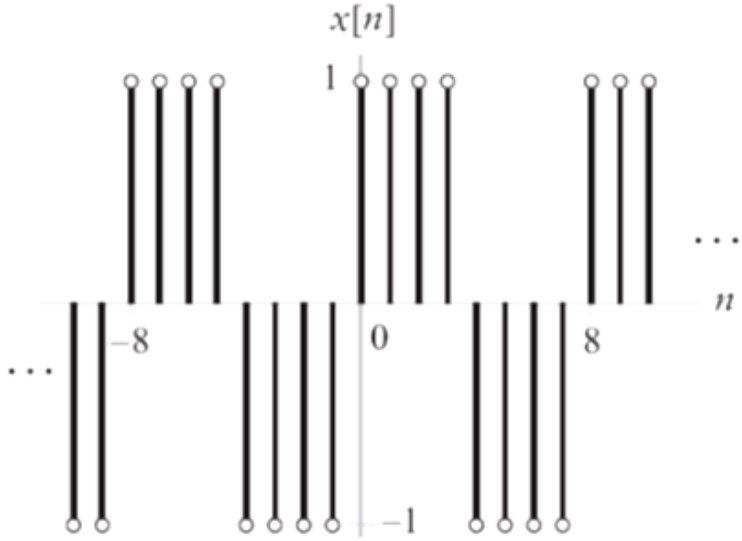


Fig. 4C

5A) (4)

Consider a signal

$$y(t) = x(t)cos(3\pi t),$$

Where the spectrum of the signal *x(t)* is shown in the figure 5A. Sample the signal with sampling rate $\omega_s = 9\pi$. Sketch the magnitude spectrum of both $y(t)$ and its sampled signal. Also, sketch the frequency response of the filter required for ideal reconstruction of $y(t)$.

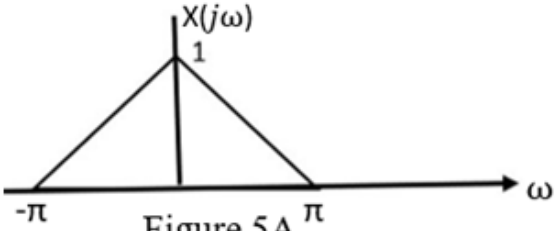


Figure 5A

5B) Using properties, find the Fourier representation of (3)

$$x(n) = n \ sin\left(\frac{\pi}{3}n\right)\left(\frac{1}{2}\right) \quad u[n] * \delta\ (n-6)$$

5C) The system function H(*z*) is given by: (3)

$$H(z) = \frac{3-4\,z^{-1}}{1-3.5\,z^{-1}+1.5\,z^{-2}}$$

Determine the impulse response for the following conditions:

i) Stable system

ii) Causal system

Can the system be both causal and stable? Justify.

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